



ALLIANCE
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NAAC
GRADE A+
ACCREDITED UNIVERSITY

Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 7

Title: Developing finance KPI dashboards

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Problem Statement

The organization has a large amount of financial transaction data containing information about budgets, actual spending, departments, categories, regions and payment methods. However, raw financial data is difficult to understand and analyze without proper visualization.

The main objective of this project is to analyze the financial transaction data using **Microsoft Power BI** and develop an interactive dashboard that provides a clear comparison between the **Budget Amount and Actual Amount**. The dashboard also helps in understanding spending patterns across different departments, categories, regions and payment methods.

Methodology / Steps Followed

Step 1: Data Collection

- Obtained the **Finance Transaction dataset** containing financial transaction records.
- The dataset contains **10,010 transaction records**.
- Each transaction contains information related to the date, department, category, region, budget amount, actual amount, payment method and transaction ID.
- The dataset was used to understand the organization's overall spending and budget performance.

Step 2: Data Import

- Opened **Microsoft Power BI Desktop**.
- Imported the finance dataset using the **Get Data** option.
- Loaded the dataset into Power BI for further analysis and visualization.
- Verified that all the required columns were successfully imported into Power BI.

Step 3: Data Cleaning and Transformation

- Checked the dataset for missing, duplicate or inconsistent values.
- Verified the data types of all columns.
- Converted the **Date** column into the appropriate date format.
- Checked the **Budget Amount** and **Actual Amount** columns to ensure that they were stored as numerical values.

- Reviewed categorical columns such as **Department, Category, Region and Payment Method**.
- Verified the **Transaction ID** field for identifying individual transactions.
- Removed or ignored unnecessary fields that were not required for the dashboard.
- Ensured that the data was properly prepared before creating the DAX measures and visualizations.

OUTPUT:

	Date	A ^B _C Department	A ^B _C Category	A ^B _C Region	i ² ₃ Budget Amount	i ² ₃ Actual Amount	A ^B _C Payment
	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty
1	11-05-2023	Sales	Travel	North	126096	43048	Card
2	11-11-2023	Marketing	Salaries	East	19702	87896	Bank Tra
3	02-05-2021	IT	Training	Central	108523	103632	Card
4	12-04-2022	Marketing	Salaries	North	114711	105574	Cash
5	27-11-2021	Sales	Utilities	Central	121895	64314	UPI
6	29-03-2021	Finance	Marketing	East	37762	77044	Bank Tra
7	22-05-2023	IT	Marketing	West	144311	69127	UPI
8	11-05-2021	HR	Marketing	Central	22567	53015	Cash
9	09-02-2023	IT	Infrastructure	West	95737	58734	UPI
10	10-12-2021	IT	Salaries	East	16857	153432	Cash
11	17-03-2023	IT	Utilities	East	75603	73056	Bank Tra
12	21-01-2022	Finance	Utilities	Central	85399	109122	Cash
13	14-08-2023	Sales	Marketing	Central	68086	70502	Bank Tra
14	04-10-2021	HR	Training	East	99145	51102	UPI
15	05-04-2022	Sales	Infrastructure	Central	42270	148677	Cash
16	22-01-2021	Marketing	Infrastructure	Central	130071	79899	Card
17	10-09-2021	Marketing	Marketing	Central	12012	99017	Card
18	18-01-2023	Marketing	Marketing	Central	69049	120109	Card
19	07-05-2023	Sales	Salaries	North	94989	30133	Bank Tra
20	20-04-2022	Operations	Training	East	99586	30720	Cash

Step 4: Data Modeling

- Created DAX measures to calculate important financial KPIs.
- Created measures for **Total Budget** and **Total Actual Amount**.
- Created a **Budget Variance** measure to compare actual expenditure with the allocated budget.
- Created measures for **Average Budget** and **Average Actual Amount**.
- Created a measure for **Total Transactions**.
- These measures were used in KPI cards, charts and other dashboard visualizations.

DAX Screenshots:

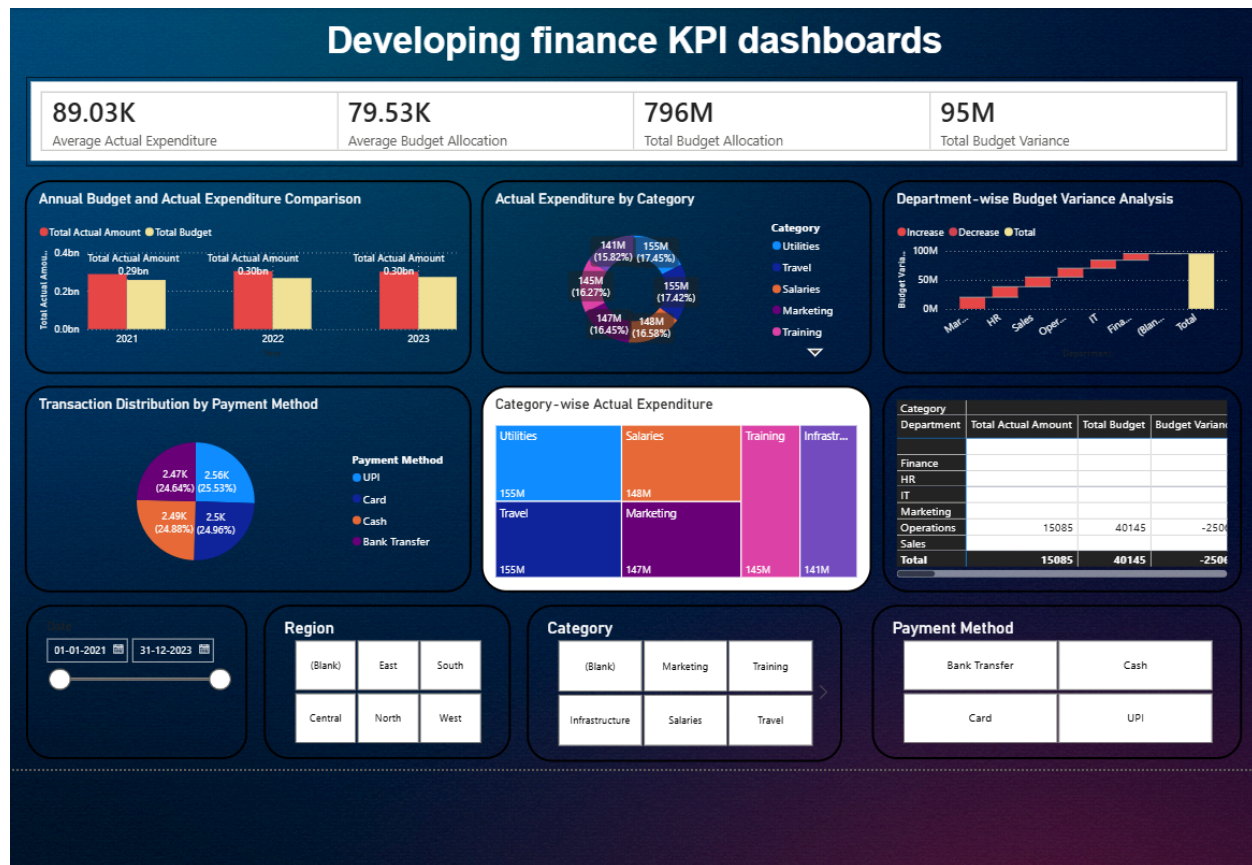
1 Average Actual Amount = AVERAGE(Sheet1[Actual Amount])

- ```
1 Average Budget = AVERAGE(Sheet1[Budget Amount])
```
- ```
1 Budget Variance = [Total Actual Amount] - [Total Budget]
```
- ```
1 Total Actual Amount = SUM(Sheet1[Actual Amount])
```
- ```
1 Total Transactions = DISTINCTCOUNT('Sheet1'[Transaction ID])
```
- ```
1 Total Budget = SUM('Sheet1'[Budget Amount])
```

### Step 5: Dashboard Development

- Created KPI cards to display important financial metrics such as **Average Actual Amount**, **Average Budget**, **Total Budget** and **Budget Variance**.
- Developed a **Clustered Column Chart** to compare Total Actual Amount and Total Budget across different years.
- Created a **Donut Chart** to show the distribution of Actual Amount across different categories.
- Developed a **Waterfall Chart** to analyze Budget Variance across different departments.
- Created a **Pie Chart** to analyze transactions based on different payment methods.
- Developed a **Treemap** to represent category-wise Actual Amount and identify categories with higher spending.
- Created a **Matrix** to compare Total Actual Amount, Total Budget and Budget Variance across departments.
- Added interactive slicers for **Date**, **Region**, **Category** and **Payment Method**.
- Designed an interactive Finance KPI Dashboard that allows users to analyze financial performance from different perspectives.

## Output



The final Power BI dashboard provides:

- Overall Budget and Actual Expenditure Analysis
- Annual Budget and Actual Expenditure Comparison
- Category-wise Actual Expenditure Analysis
- Department-wise Budget Variance Analysis
- Transaction Distribution by Payment Method
- Department-wise Financial Performance
- Interactive Filters and Slicers

### Step 6: Dashboard Development

- Analyzed the overall relationship between **Budget Amount and Actual Amount**.
- Identified departments with higher or lower budget variance.
- Compared spending across different categories such as Travel, Utilities, Salaries, Marketing, Training and Infrastructure.

- Analyzed the distribution of transactions across different payment methods.
- Compared financial performance across different regions.
- Studied the yearly trend of budget and actual expenditure.
- Used interactive slicers to perform focused analysis based on date, region, category and payment method.
- Derived useful financial insights that can help in budget monitoring and expenditure management.

## Conclusion

The **Finance KPI Dashboard using Power BI** successfully converts raw financial transaction data into meaningful and interactive visual insights.

The dashboard makes it easier to compare **Budget Amount with Actual Amount**, identify budget variances and understand spending patterns across departments, categories, regions and payment methods.

The use of **DAX measures, KPI cards, charts, matrix visualizations and interactive slicers** provides management with a clear overview of financial performance.

Overall, the project demonstrates how Power BI can be used to analyze financial data effectively and support **budget monitoring, expenditure analysis and data-driven financial decision-making**.

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GITHUB LINK :

<https://github.com/MaharshiGoswami/Building-Sales-Performance-Dashboard-using-Power-BI/blob/main/BudgetVsActual%20Analysis.pbix>